



instructables

Homemade Portable Bluetooth Speaker



by stan9801

Hey guys,

I made this bluetooth speaker last year, It took me about 3 months to build it.
I used only quality components for this speaker.
Here is a video about it!

<https://www.youtube.com/watch?v=sV9X4A-DNSw>

I hope you guys like my speaker!

<https://www.youtube.com/watch?v=sV9X4A-DNSw>





Step 1: Design Your Speaker and Collect All the Parts

I made my speaker design using Google Sketchup. I made sure the volume was a little bit bigger than the manufacturer advices for the speakers i am using , because i also have to fit all the other components inside.

First i searched the manufacturers website for the correct volume.

When i finished my design, i started collect all the parts.

Parts list:

Speakers/tweeters:

<http://www.daytonaudio.com/index.php/ds135-8-5-des...>

<http://www.daytonaudio.com/index.php/loudspeaker-c...> Audio amp:

(I use this board with the lithium battery extension board)

<http://store3.sure-electronics.com/2-x-30-watt-cla...>

Batteries:

<http://www.ebay.com/itm/4x-18650-6000mAh-4-2V-Li-i...> Charger:

I am using a 19v charger recommended by the seller.

<http://store3.sure-electronics.com/ps-sp11506>

Crossover:

I used a crossover filter set from a dutch store.

<http://store.accusafe.nl/catalogsearch/result/?q=f..>

Case feet:

<http://www.ebay.com/itm/4PCS-Aluminum-Isolation-St...> Volume knob:

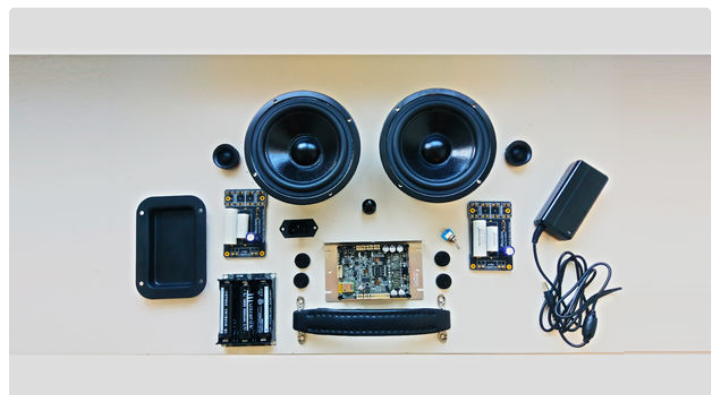
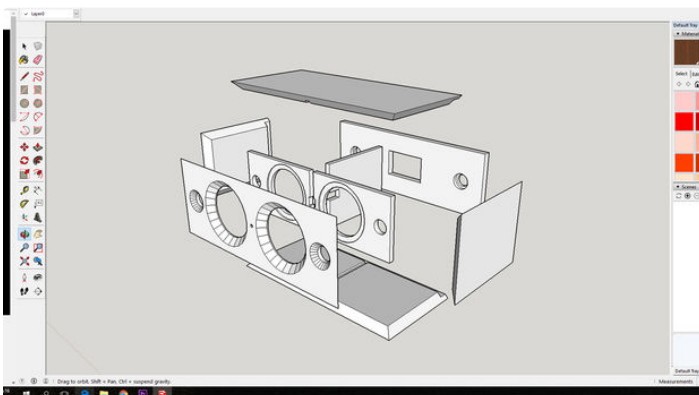
<http://www.ebay.com/itm/2-PCS-Volume-Cap-Knob-Elec...> Leather handle:

http://www.ebay.com/sch/i.html?_odkw=leather+strap...

Metal plate for the back:

<http://store.accusafe.nl/inbouw-schotel-zwart-130x...>

For the enclosure i had some oak wood laying around.





Step 2: Building the Enclosure

I made my speaker completely out of 18mm oak wood

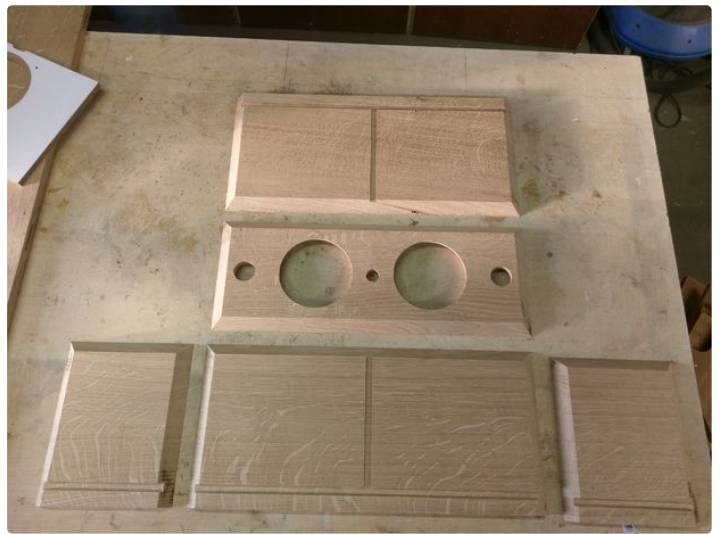
First i changed my oak log planks into an 18mm oak wood planks.

When that was done i was able to make use of an CNC machine to cut out al the holes for the speaker, the bass reflex tubes and the metal plate on the back.

When i got al the wood panels i glued everything together using a strong wood glue.

I also stained the whole speaker for the shiny finish.





Step 3: Install All the Components.

For the speaker mounting i used 2 10mm pieces of wood to fix the speakers in place.

(i didnt make a good picture of it but you can see it on one of the pictures.)

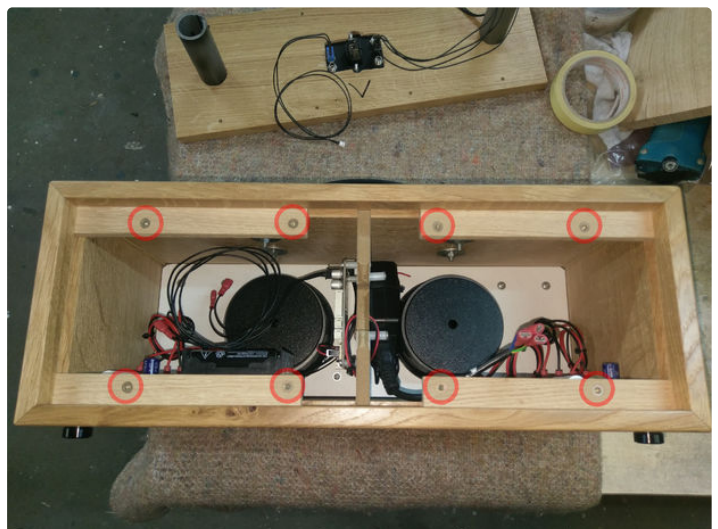
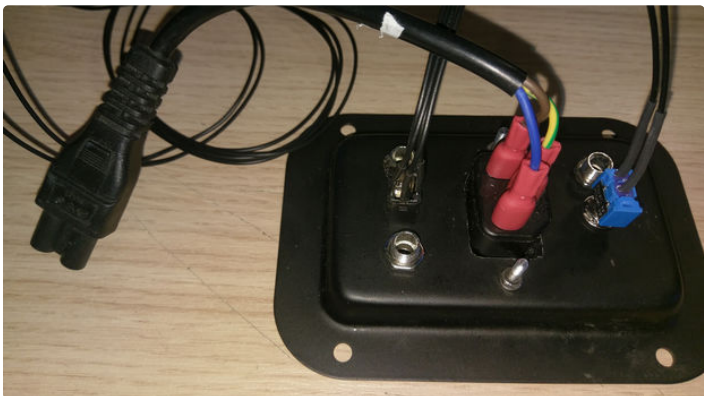
For the amplifier, crossover and battery board i used some standoffs to fix these into place.

It is possible to pull out the complete mid piece of

wood where my charger and amp are installed on.

To close my speaker with the backpiece i installed some "things"(See picture) where i can screw a bolt into.

After this the speaker is done! I really hope you guys like my speaker.





Woow it really looks amazing. Great work man :)



Insanely beautiful <3



Looks truly amazing.



That looks really nice :)